

BBM 448 - Assignment 1

Extending the Commit Log Service in Go

Context

In DSwG – Chapter 1, we studied a simple **commit log service** in Go that supports:

- `POST /` → append a record to the log
- `GET /` → read a record by offset

Now you will **extend this service** by adding two new features. This assignment is designed to help you:

- Understand REST API design
- Work with JSON over HTTP
- Extend existing Go code safely and correctly



Task 1 – List All Records

Implement a new endpoint:

- **Endpoint:** `GET /records`
- **Behavior:** Return **all log records** in a JSON array.
- **Thread-safety:** Ensure you lock/unlock the log as needed.



Task 2 – Clear All Records

Implement another endpoint:

- **Endpoint:** `DELETE /records`
 - **Behavior:** Clear the entire log (reset to empty).
 - **Thread-safety:** Lock/unlock when modifying the log.
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Expected Behavior & Test Commands

Start your server as usual:

```
go run cmd/server/main.go
```

1. Append three records

```
curl -X POST localhost:8080 -d '{"record": {"value": "SGVsbG8="}}'
curl -X POST localhost:8080 -d '{"record": {"value": "V29ybGQ="}}'
curl -X POST localhost:8080 -d '{"record": {"value": "R28="}}'
```

2. List all records

```
curl -X GET localhost:8080/records
```

Expected Output (JSON):

```
{
  "records": [
    {"value": "SGVsbG8=", "offset": 0},
    {"value": "V29ybGQ=", "offset": 1},
    {"value": "R28=", "offset": 2}
  ]
}
```

3. Clear all records

```
curl -X DELETE localhost:8080/records
```

Expected Output:

Status code 200 OK and no body.

4. List after clearing

```
curl -X GET localhost:8080/records
```

Expected Output:

```
{
  "records": []
}
```

Submission Guidelines

- Submit your updated `http.go` file to `submit.cs.hacettepe.edu.tr`
- Deadline: 23.59 @ Monday, 13.10.2025 (no late submission, no extension)